MATH 10A — STRUCTURE AND CONCEPTS IN MATH I

§1.2
The Egyptian Numeration System

Chapter 1 — Numeration Systems

Today's Goal

After completing this section, you should be able to:

- Understand how an additive numeration system works.
- Read and write numbers using Egyptian numerals.

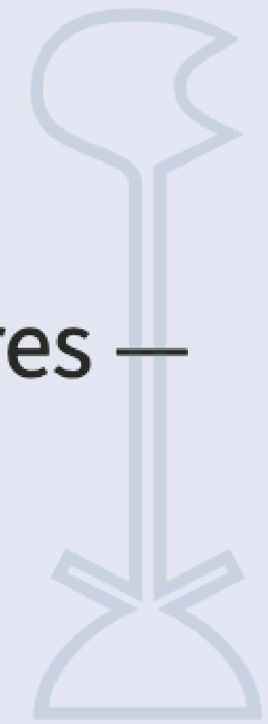
LT.2 Numeration Systems

I can convert numbers between various numeration systems including Roman, Egyptian, Mayan, Babylonian, and Hindu-Arabic.

Before There Were Digits

How would you write "one hundred" if the digits 0–9 had never been invented?

Five thousand years ago, Egyptian scribes answered that question with pictures — just seven of them.



The Seven Egyptian Glyphs

Egyptian Glyph	Description	Number
𓏺	staff	1
𓎆	heel bone	10
𓍢	spiral	100
𓆼	lotus flower	1,000
𓂭	finger	10,000
𓆐	tadpole	100,000
𓁨	astonished person	1,000,000

Try It Before Any Rules

How would you write 400 with only these seven symbols?

𓍢𓍢𓍢𓍢

Four spirals, each worth 100 — together, 400. You just used the system's only rule.

One Rule Makes It Work

The Egyptian numeration system is an example of an *additive numeration system*.

Definition 1.2.2

An *additive numeration system* is a numeration system in which the value of a set of glyphs is the sum of the values of each glyph in the set.

The order of the glyphs helps humans read — but only the *sum* carries the meaning.

Reading by Counting and Adding

What number is represented by the Egyptian numeral shown below?

𓆼𓆼𓍢𓍢𓍢𓍢𓍢𓎆𓎆𓎆𓏺𓏺𓏺𓏺

Egyptian numerals are customarily written in decreasing order from left to right.

Solution. Count each glyph, multiply by its value, and add:

Egyptian Glyph	Count	Total Value
𓆼	2	$2 \cdot 1{,}000 = 2{,}000$
𓍢	5	$5 \cdot 100 = 500$
𓎆	3	$3 \cdot 10 = 30$
𓏺	4	$4 \cdot 1 = 4$

Adding these gives us the Hindu-Arabic numeral $2{,}534$.

What We Just Did, Formalized

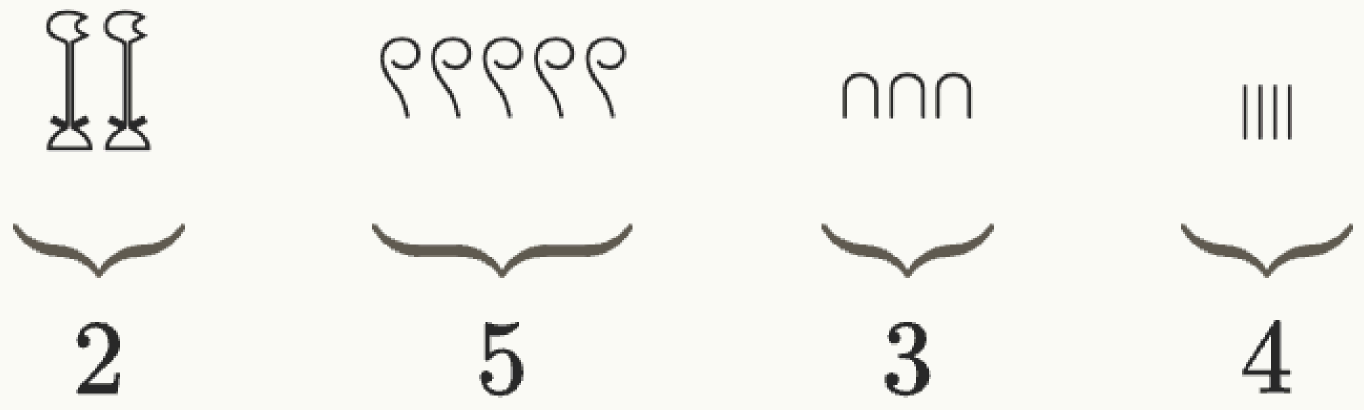
Procedure 1.2.3. Converting an Egyptian Numeral to the Corresponding Hindu-Arabic Numeral

To convert an Egyptian numeral to the corresponding Hindu-Arabic numeral, do the following.

1. Identify which Egyptian glyphs (see Table 1.2.1) are being used.
2. Count the number of each different Egyptian glyph that is being used.
3. For each different Egyptian glyph used, multiply the value of the Egyptian glyph by how many of them were used.
4. Add all of the subtotals from Step 3.

A Tempting Shortcut

A shortcut suggests itself: *"The glyphs are already listed in decreasing order — so just count how many of each there are:"*



"Therefore, the number is $2{,}534$*."*

It worked this time. What could go wrong?

Putting the Shortcut to the Test

What number is represented by this Egyptian numeral? Try the shortcut, then Procedure 1.2.3.

Solution. The shortcut first: count how many of each different glyph appear, reading left to right.

fingers: 4 spirals: 2 heel bones: 6 staffs: 3

Put those counts side by side and the shortcut reads the numeral as $4{,}263$.

Checking with Procedure 1.2.3

The shortcut read this numeral as $4{,}263$. Time to check its work.

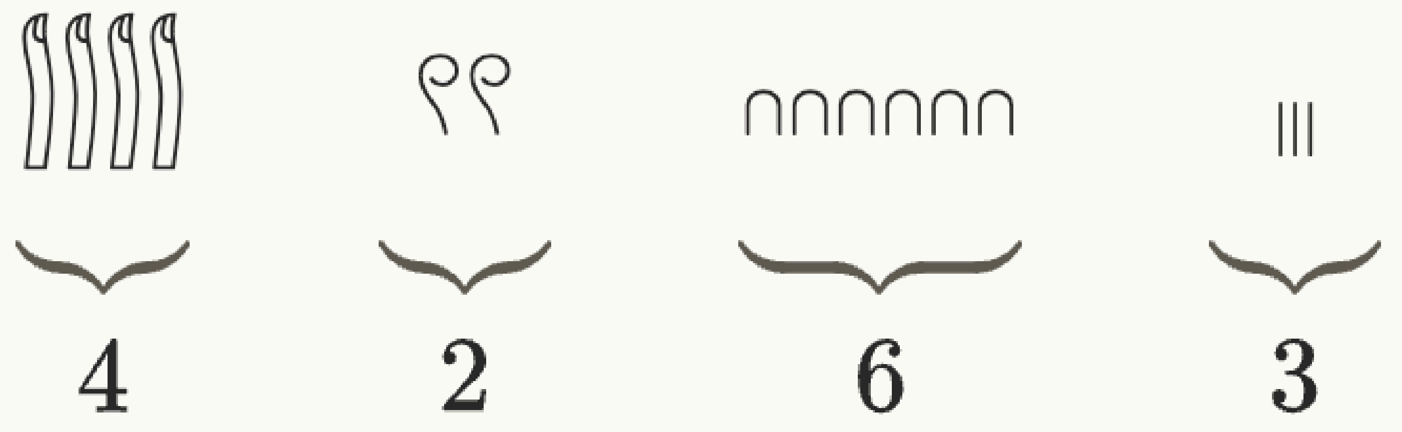
Solution, continued. Count each glyph, multiply by its value, and add:

Egyptian Glyph	Count	Total Value
𓆼	4	$4 \cdot 10{,}000 = 40{,}000$
𓍢	2	$2 \cdot 100 = 200$
𓎆	6	$6 \cdot 10 = 60$
𓏺	3	$3 \cdot 1 = 3$

Adding gives $40{,}263$ — but the shortcut said $4{,}263$.

Why the Shortcut Fails

The shortcut counts each different glyph and strings the counts together:



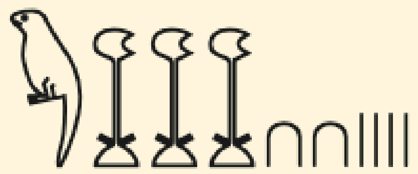
Read as digits, the counts say "$4{,}263$" — but the numeral is $40{,}263$.

The counts $4, 2, 6, 3$ say nothing about each glyph's *value* — four fingers are worth $40{,}000$, and no glyph marks the empty thousands place. Ignoring the value of each glyph that is used will cause problems.

Now You Try: Reading

You Try 1.2.1.

What number does this Egyptian numeral represent?



Answer. $100{,}000 + 3{,}000 + 20 + 4 = 103{,}024$. The glyphs that do not appear contribute 0.

Writing Goes Through Expanded Form

Procedure 1.2.8. Converting a Hindu-Arabic Numeral to the Corresponding Egyptian Numeral

To convert a Hindu-Arabic numeral to the corresponding Egyptian numeral, do the following.

1. Write the number in expanded form.
2. Use the expanded form to determine how many of each Egyptian glyph is needed.

Expanded Form Does the Work

Write the number $24{,}153$ as an Egyptian numeral.

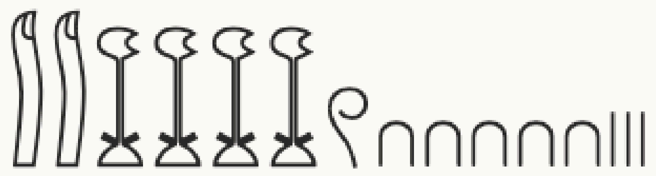
Solution. We begin by writing $24{,}153$ in expanded form.

$$24{,}153 = 20{,}000 + 4{,}000 + 100 + 50 + 3.$$

Part	Egyptian Glyph
$20{,}000 = 2 \cdot 10{,}000$	𓂭𓂭
$4{,}000 = 4 \cdot 1{,}000$	𓆼𓆼𓆼𓆼
$100 = 1 \cdot 100$	𓍢
$50 = 5 \cdot 10$	𓎆𓎆𓎆𓎆𓎆
$3 = 3 \cdot 1$	𓏺𓏺𓏺

Assembling the Numeral

Each part of the expanded form became one glyph group. Combining these gives us the Egyptian numeral



Reading the numeral back, each glyph group matches one part of the expanded form: $20{,}000 + 4{,}000 + 100 + 50 + 3 = 24{,}153.$

Now You Try: Writing

You Try 1.2.2.

Write $20{,}507$ as an Egyptian numeral.

Answer. Two fingers, five spirals, seven staffs — no lotus flowers or heel bones needed: